Logarithmic and Exponential Functions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Logarithmic & Exponential Functions Practice Exercises**

[Exponential, Logarithmic & Logistic Dynamics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

2 **[Foundational]** • **Target: Logarithmic & Exponential Functions Practice Exercises**

Solve the logarithmic equation $\log_3(x + 6) + \log_3(x) = 3$ and eliminate any extraneous solutions.

Combine the logarithms on the left-hand side using the product rule: $\log_3[x(x + 6)] = 3$. Convert to exponential form: $x(x + 6) = 3^3 = 27 \Rightarrow x^2 + 6x - 27 = 0$. Factor the quadratic: $(x + 9)(x - 3) = 0 \Rightarrow x = -9$ or $x = 3$. Check the domain restrictions in the original equation:For $x = -9$: $\log_3(-9)$ is undefined, so $x = -9$ is an extraneous solution.For $x = 3$: $\log_3(9) + \log_3(3) = 2 + 1 = 3$. Valid ✓.Therefore, the unique solution is $x = 3$.

3 **[Foundational]** • **Target: Logarithmic & Exponential Functions Practice Exercises**

A radioactive substance follows the decay model $M(t) = M_0 e^{-kt}$. If 20% of the sample decays in 15 days, find the exact decay constant k and the substance's half-life in days to the nearest tenth.

If 20% decays in 15 days, then 80% remains at $t = 15$: $0.80M_0 = M_0 e^{-15k} \Rightarrow e^{-15k} = 0.80$. Take the natural logarithm of both sides: $-15k = \ln(0.80) \Rightarrow k = -\frac{\ln(0.80)}{15} = \frac{\ln(1.25)}{15} \approx 0.01488 \text{ day}^{-1}$. The half-life $t_{1/2}$ satisfies $e^{-kt_{1/2}} = 0.5$: $t_{1/2} = \frac{\ln 2}{k} = \frac{15\ln 2}{-\ln(0.80)} \approx \frac{15(0.6931)}{0.2231} \approx 46.6$ days.

4 **[Foundational]** • **Target: Logarithmic & Exponential Functions Practice Exercises**

Solve the logarithmic equation $\log_2(x + 3) + \log_2(x - 3) = 4$ for all real solutions x. State any extraneous roots.

Step 1: State domain restrictions: $x + 3 > 0 \Rightarrow x > -3$ and $x - 3 > 0 \Rightarrow x > 3$, so $x > 3$. Step 2: Combine logarithms: $\log_2[(x + 3)(x - 3)] = 4 \Rightarrow \log_2(x^2 - 9) = 4$. $x^2 - 9 = 2^4 = 16 \Rightarrow x^2 = 25 \Rightarrow x = \pm 5$. Step 3: Eliminate extraneous solutions: $x = -5$ violates $x > 3$ (producing negative arguments), so it is extraneous. The unique valid solution is $x = 5$.

5 **[Foundational]** • **Target: Logarithmic & Exponential Functions Practice Exercises**

If $f(x) = 4x - 3$ and $g(x) = x^2 + 1$, find (a) $gf(2)$ and (b) an expression for $g^{-1}(x)$ for $x \geq 1$.

(a) Inner function first: $f(2) = 4(2) - 3 = 5$. Then $g(5) = 5^2 + 1 = 26$. (b) Inverse of g: Let $y = x^2 + 1 \Rightarrow x^2 = y - 1 \Rightarrow x = \sqrt{y - 1}$. Thus $g^{-1}(x) = \sqrt{x - 1}$ for $x \geq 1$.

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6 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

[Probability & Statistics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

7 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

[Transformations & Vectors · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Logarithmic & Exponential Functions Practice Exercises

[Unit Circle & Trigonometric Identities · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

9 [Intermediate] • Target: Logarithmic & Exponential Functions Practice Exercises

[Conic Sections & Analytic Geometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

10 [Intermediate] • Target: Logarithmic & Exponential Functions Practice Exercises

[Polar Coordinates, Complex Numbers & De Moivre's Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

11 [Intermediate] • Target: Logarithmic & Exponential Functions Practice Exercises

[Sequences, Series, Induction & the Binomial Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

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12 **[Intermediate]** • **Target: Logarithmic & Exponential Functions Practice Exercises**

[Trigonometry & Triangle Laws · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

13 **[Intermediate]** • **Target: Logarithmic & Exponential Functions Practice Exercises**

[Vectors, Lines & Parametric Kinematics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

14 **[Intermediate]** • **Target: Logarithmic & Exponential Functions Practice Exercises**

In ΔABC, $a = 7$, $b = 9$, and $c = 10$. Find the measure of the smallest angle to the nearest tenth of a degree, and compute the area of ΔABC using Heron's formula.

Step 1: Identify smallest angle: The smallest angle is A, opposite the shortest side $a = 7$. Step 2: Law of Cosines for angle A: $\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{9^2 + 10^2 - 7^2}{2(9)(10)} = \frac{81 + 100 - 49}{180} = \frac{132}{180} = \frac{11}{15} \approx 0.7333$. $A = \arccos(11/15) \approx 42.8^\circ$. Step 3: Heron's formula for area: Semi-perimeter $s = \frac{7 + 9 + 10}{2} = 13$..

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Logarithmic & Exponential Functions Practice Exercises** • **Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure**

If $\sin\theta = \frac{3}{5}$ with θ terminating in Quadrant I, find the exact values of $\cos\theta$, $\tan\theta$, and $\sin(2\theta)$.

Step 1: Compute $\cos\theta$ via Pythagorean identity: $\cos\theta = \sqrt{1 - \sin^2\theta} = \sqrt{1 - \frac{9}{25}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$. Step 2: Compute $\tan\theta$: $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{3/5}{4/5} = \frac{3}{4}$. Step 3: Compute $\sin(2\theta)$ using the double-angle identity: $\sin(2\theta) = 2\sin\theta\cos\theta = 2\left(\frac{3}{5}\right)\left(\frac{4}{5}\right) = \frac{24}{25}$.

16 **[Challenging]** • **Target: Logarithmic & Exponential Functions Practice Exercises** • **Cross-Lesson: 6.01 Permutations and Combinations**

Find the infinite sum of the geometric series $12 + 8 + \frac{16}{3} + \frac{32}{9} + \ldots$. Verify the condition for convergence.

Step 1: Determine first term and common ratio: First term $a_1 = 12$. Common ratio $r = \frac{8}{12} = \frac{2}{3}$. Step 2: Check convergence condition: Since $|r| = \left|\frac{2}{3}\right| = \frac{2}{3} < 1$, the infinite series converges. Step 3: Compute infinite sum: $S_\infty = \frac{a_1}{1 - r} = \frac{12}{1 - 2/3} = \frac{12}{1/3} = 36$.

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17 **[Challenging]** • **Target: Logarithmic & Exponential Functions Practice Exercises** • **Cross-Lesson: 7.01 Differentiation and Rates of Change**

The diagram shows a sector of a circle with radius r. The angle at the centre is θ radians. If the arc length is USD $6 cm and the radius is USD 3 cm, find the angle θ in radians.

The formula relating arc length (s), radius (r), and angle (θ) in radians is $s = r\theta$. We are given $s = 6$ cm and $r = 3$ cm. Substituting these values into the formula, we get $6 = 3\theta$. To find θ, we divide both sides by 3: $\theta = \frac{6}{3} = 2$ radians. Therefore, the angle at the centre is USD 2 radians.

18 **[Challenging]** • **Target: Logarithmic & Exponential Functions Practice Exercises** • **Cross-Lesson: 8.01 Integration and Kinematics**

Convert the polar equation $r = 4\cos\theta$ to Cartesian rectangular form, classify the curve, and find its center and radius.

Step 1: Multiply both sides by r: $r^2 = 4r\cos\theta$. Step 2: Substitute $r^2 = x^2 + y^2$ and $r\cos\theta = x$: $x^2 + y^2 = 4x \Rightarrow x^2 - 4x + y^2 = 0$. Step 3: Complete the square on x: $(x^2 - 4x + 4) + y^2 = 4 \Rightarrow (x-2)^2 + y^2 = 4$. This represents a circle with center $(2, 0)$ and radius $r = 2$.

19 **[Challenging]** • **Target: Logarithmic & Exponential Functions Practice Exercises** • **Cross-Lesson: 1.01 Functions, Modulus and Quadratics**

Convert the polar coordinate point $P\left(6, \frac{4\pi}{3}\right)$ to exact Cartesian coordinates (x, y).

Step 1: Apply conversion formulas: $x = r\cos\theta = 6\cos\left(\frac{4\pi}{3}\right) = 6\left(-\frac{1}{2}\right) = -3$. $y = r\sin\theta = 6\sin\left(\frac{4\pi}{3}\right) = 6\left(-\frac{\sqrt{3}}{2}\right) = -3\sqrt{3}$. The Cartesian coordinates are $(-3, -3\sqrt{3})$.

20 **[Challenging]** • **Target: Logarithmic & Exponential Functions Practice Exercises** • **Cross-Lesson: 2.01 Simultaneous Equations and Inequalities**

Find the center, radius, and standard Cartesian equation of the circle given by the general equation $x^2 + y^2 - 8x + 6y - 11 = 0$.

Step 1: Group variables and complete the square: $(x^2 - 8x) + (y^2 + 6y) = 11$. $(x^2 - 8x + 16) + (y^2 + 6y + 9) = 11 + 16 + 9 = 36$. $(x-4)^2 + (y+3)^2 = 36$. Step 2: Extract circle invariants: Center $(h, k) = (4, -3)$ and radius $r = \sqrt{36} = 6$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Solve $7^x = 100$ for x, expressed using logarithms.

a. $x = \frac{100}{7}$

b. $x = 100 - 7$

c. $x = \log_7(100) = \frac{\ln 100}{\ln 7}$ ✓

d. $x = \log_{100}(7)$

Take $\ln$ of both sides: $x \ln 7 = \ln 100$, so $x = \frac{\ln 100}{\ln 7} \approx 2.367$.

22 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

A population of 500 grows at 5% per year, modeled by $P(t) = 500(1 + 0.05)^t$. Approximately how many years until the population reaches $1.5\times$ its initial size (i.e. $P = 1.5 \cdot 500$)?

a. **8.3 years** ✓

b. 10.8 years

c. 6.3 years

d. 10.3 years

Solve $1.5 = (1 + 0.05)^t$: $t = \frac{\ln 1.5}{\ln(1 + 0.05)} \approx 8.3$ years.

23 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Write $5^4 = 625$ in logarithmic form.

a. $\log_{625}(5) = 4$

b. $\log_5(4) = 625$

c. $\log_4(5) = 625$

d. $\log_5(625) = 4$ ✓

$b^x = y \Leftrightarrow \log_b(y) = x$, so $\log_5(625) = 4$.

24 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Expand $\log\left(\frac{x}{y}\right)$ using properties of logarithms. (variant 6)

a. $\frac{\log x}{\log y}$

b. $\log(x) \cdot \log(y)$

c. $\log x + \log y$

d. $\log x - \log y$ ✓

Quotient rule: $\log(x/y) = \log x - \log y$.

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25 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Expand $\log\left(\frac{x^2}{y^3}\right)$ using properties of logarithms.

a. $\log(x) \cdot \log(y)$

b. $2\log x - 3\log y$ ✓

c. $2\log x + 3\log y$

d. $\frac{\log x}{\log y}$

Combine power and quotient rules.

26 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Evaluate the definite integral: $\int_0^1 (2x+1)^3\,dx$.

a. $\frac{80}{2}$

b. $\frac{84}{8}$

c. $\frac{80}{4}$

d. $\frac{80}{8}$ ✓

Substitute $u = 2x + 1 \Rightarrow du = 2\,dx$. Integral: $\frac{1}{2}\left[\frac{u^4}{4}\right]_1^3 = \frac{(3)^4 - 1^4}{8} = \frac{80}{8}$.

27 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Find the number of distinct permutations of all letters in STATISTICS if all 3 'S's must stay together as a single block.

a. **3360** ✓

b. 1680

c. 6720

d. 50400

Treat SSS as 1 item. Items: [SSS], T, A, T, I, T, I, C (8 items, 3 T's, 2 I's): 8!/(3!2!) = 3360.

28 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Find the number of distinct permutations of all letters in COMMITTEE if the 2 'M's must stay together.

a. **10080** ✓

b. 20160

c. 45360

d. 5040

Treat MM as 1 item. Items: [MM], C, O, I, T, T, E, E (8 items, 2 T's, 2 E's): 8!/(2!2!) = 10080.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Logarithmic & Exponential Functions Practice Exercises

A committee of 4 people is to be chosen from 5 men and 4 women. How many committees consist of exactly 2 men and 2 women?

a. 60 ✓

b. 120

c. 55

d. 36

$\binom{5}{2} \times \binom{4}{2} = \frac{5 \times 4}{2} \times \frac{4 \times 3}{2} = 60$.

30 [Intermediate] • Target: Logarithmic & Exponential Functions Practice Exercises

Find the number of distinct permutations of all letters in ADDITION if both 'D's must stay together.

a. 10080

b. 5040

c. 2520 ✓

d. 1260

Treat DD as 1 item. Items: [DD], A, I, I, T, O, N (7 items, 2 I's): 7!/2! = 2520.

31 [Intermediate] • Target: Logarithmic & Exponential Functions Practice Exercises

Find the number of distinct permutations of all letters in ARRANGE if the 2 'R's must be together.

a. 180

b. 360 ✓

c. 720

d. 1260

Treat RR as 1 item. Items: [RR], A, A, N, G, E (6 items, 2 A's): 6!/2! = 360.

32 [Intermediate] • Target: Logarithmic & Exponential Functions Practice Exercises

Find the number of distinct permutations of all letters in SUCCESS if all 3 'S's must stay together.

a. 30

b. 420

c. 120

d. 60 ✓

Treat SSS as 1 item. Items: [SSS], U, C, C, E (5 items, 2 C's): 5!/2! = 60.

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33 [Intermediate] • Target: Logarithmic & Exponential Functions Practice Exercises

Find the number of distinct permutations of all letters in PARALLEL if all 3 'L's must stay together as a single block.

a. 180

b. 720

c. **360** ✓

d. 3360

Treat LLL as 1 item. Remaining items: P, A, R, A, E. 6!/2! = 360.

34 [Intermediate] • Target: Logarithmic & Exponential Functions Practice Exercises

The graph of the function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of the parameter a.

a. 1

b. **2** ✓

c. 3

d. 4

We are given the function $f(x) = a \cdot b^x$ and two points it passes through: $(1, 12)$ and $(3, 108)$. Using the first point $(1, 12)$, we have: $12 = a \cdot b^1$ (Equation 1) Using the second point $(3, 108)$, we have: $108 = a \cdot b^3$ (Equation 2) To find a and b, we can divide Equation 2 by Equation 1: $\frac{108}{12} = \frac{a \cdot b^3}{a \cdot b^1}$ $9 = b^2$ Taking the square root of both sides, and assuming $b > 0$ for exponential functions: $b = 3$ Now, substitute the value of $b = 3$ back into E.

35 [Intermediate] • Target: Logarithmic & Exponential Functions Practice Exercises

A population of bacteria, initially P_0 cells, grows according to the logistic model $P(t) = \frac{L}{1 + Ae^{-kt}}$, where t is in hours, L is the carrying capacity, and A and k are positive constants. At $t = 0$, the population is $P(0) = \frac{L}{3}$. After 2 hours, the population has doubled to $P(2) = \frac{2L}{3}$. Find the exact value of e^{2k} in terms of L and P_0.

a. $\frac{9}{4}$ ✓

b. $\frac{4}{9}$

c. 3

d. $\frac{3}{2}$

The logistic model is given by $P(t) = \frac{L}{1 + Ae^{-kt}}$. At $t = 0$, $P(0) = \frac{L}{3}$. Substituting into the model: $\frac{L}{3} = \frac{L}{1 + Ae^{-k \cdot 0}}$ $\frac{L}{3} = \frac{L}{1 + A}$ $1 + A = 3$ $A = 2$. So the model becomes $P(t) = \frac{L}{1 + 2e^{-kt}}$. We are also given that $P(2) = \frac{2L}{3}$. Substituting into the model: $\frac{2L}{3} = \frac{L}{1 + 2e^{-2k}}$ $\frac{2}{3} = \frac{1}{1 + 2e^{-2k}}$ $2(1 + 2e^{-2k}) = 3$ $2 + 4e^{-2k} = 3$ $4e^{-2k} = 1$ $e^{-2k} = \frac{1}{4}$. We need.

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36 [Intermediate] • Target: Logarithmic & Exponential Functions Practice Exercises

Consider the exponential growth model $P(t) = P_0 e^{kt}$, where $P(t)$ is the population at time t and P_0 is the initial population. If the initial population is $P_0 = 500$ and after 10 years the population is $P(10) = 1200$, find the annual growth rate k. Round your answer to four decimal places.

a. 0.0588

b. 0.0875

c. 0.1200

d. **0.0434** ✓

We are given the exponential growth model $P(t) = P_0 e^{kt}$. We know that the initial population $P_0 = 500$ and that after 10 years, $P(10) = 1200$. We can substitute these values into the model: $1200 = 500e^{k \cdot 10}$ Divide both sides by 500: $\frac{1200}{500} = e^{10k}$ $2.4 = e^{10k}$ To solve for k, we take the natural logarithm of both sides: $\ln(2.4) = \ln(e^{10k})$ $\ln(2.4) = 10k$ Now, divide by 10: $k = \frac{\ln(2.4)}{10}$ Using a calculator, $\ln(2.4) \approx 0.8754687$. So.

37 [Intermediate] • Target: Logarithmic & Exponential Functions Practice Exercises

Consider the exponential growth model described by the function $P(t) = P_0 e^{kt}$, where $P(t)$ is the population at time t, P_0 is the initial population, and k is the growth constant. If the initial population P_0 is 500 and the population after 4 years is 2000, determine the value of the growth constant k to three decimal places.

a. **0.347** ✓

b. 0.416

c. 0.301

d. 0.477

The population model is given by $P(t) = P_0 e^{kt}$. We are given that $P_0 = 500$. We are also given that $P(4) = 2000$. Substituting these values into the equation, we get: $2000 = 500e^{k \cdot 4}$ Divide both sides by 500: $4 = e^{4k}$ Take the natural logarithm of both sides: $\ln(4) = \ln(e^{4k})$ $\ln(4) = 4k$ Solve for k: $k = \frac{\ln(4)}{4}$ Using a calculator, $\ln(4) \approx 1.38629$. $k \approx \frac{1.38629}{4}$ $k \approx 0.34657$ Rounding to three decimal places, $k \approx 0.347$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Logarithmic & Exponential Functions Practice Exercises • Cross-Lesson: 8.01 Integration and Kinematics

Consider the exponential function $f(x) = 2^x$. If $f(a) = 8$, what is the value of a?

a. 1

b. 2

c. **3** ✓

d. 4

The given function is $f(x) = 2^x$. We are given that $f(a) = 8$. Substituting a into the function, we get $2^a = 8$. To solve for a, we need to express 8 as a power of 2. Since $2^3 = 2 \times 2 \times 2 = 8$, we can rewrite the equation as $2^a = 2^3$. Since the bases are equal, the exponents must be equal. Therefore, $a = 3$.

39 [Challenging] • Target: Logarithmic & Exponential Functions Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

Consider the exponential function $f(x) = 2^x$. What is the value of $f(3)$?

a. 6

b. **8** ✓

c. 9

d. 5

The function is given by $f(x) = 2^x$. To find $f(3)$, we substitute $x = 3$ into the function: $f(3) = 2^3$. Calculating the power, $2^3 = 2 \times 2 \times 2 = 8$. Therefore, the value of $f(3)$ is 8.

40 [Challenging] • Target: Logarithmic & Exponential Functions Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

A function is defined by $f(x) = x^2 - 6x + 5$ for $x \geq 3$. \\ The graph of $y = f(x)$ is a parabola. A straight line with equation $y = k$ intersects the graph of $y = f(x)$ at exactly one point. \\ Find the possible value(s) of k.

a. $k = -4$

b. $k = 5$

c. $k = -4$ or $k = 5$ ✓

d. $k \geq 5$

The function is given by $f(x) = x^2 - 6x + 5$ for $x \geq 3$. \\ To find the vertex of the parabola, we can complete the square: $f(x) = (x^2 - 6x + 9) - 9 + 5 = (x - 3)^2 - 4$. \\ The vertex of the parabola is at $(3, -4)$. \\ Since the domain is restricted to $x \geq 3$, the graph of $y = f(x)$ starts at its vertex and extends upwards as x increases. \\ A horizontal line $y = k$ intersects the graph of $y = f(x)$ at exactly one point if: \\ 1. The line passes through the vertex. In this case, $k = -4$. \\ 2. The.

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41 [Challenging] • Target: Logarithmic & Exponential Functions Practice Exercises • Cross-Lesson: 3.01 Indices, Surds and Polynomials

The diagram shows the graph of the quadratic function $f(x) = x^2 + ax + b$. The graph passes through the points $(-1, 8)$ and $(3, 0)$. Find the values of a and b.

a. $a = -2, b = 3$ ✓

b. $a = 2, b = -3$

c. $a = -4, b = 1$

d. $a = 4, b = -1$

The function is given by $f(x) = x^2 + ax + b$. Since the graph passes through $(-1, 8)$, we have $f(-1) = 8$. Substituting $x = -1$ into the function: $(-1)^2 + a(-1) + b = 8$ $1 - a + b = 8$ $-a + b = 7$ (*) Since the graph passes through $(3, 0)$, we have $f(3) = 0$. Substituting $x = 3$ into the function: $(3)^2 + a(3) + b = 0$ $9 + 3a + b = 0$ $3a + b = -9$ (**) Now we have a system of two linear equations with two variables: 1) $-a + b = 7$ 2) $3a + b = -9$ Subtract equation (1) from equation (2).

42 [Challenging] • Target: Logarithmic & Exponential Functions Practice Exercises • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

The graph of $g(x) = f(x + 1) - 5$ is obtained from $f(x)$ by:

a. Shifting 1 units left and 5 units down

b. **Shifting 1 units right and 5 units down** ✓

c. Shifting 1 units right and 5 units up

d. Reflecting over the x-axis

Replacing x with $x - 1$ shifts horizontally by 1 (right if positive); adding -5 shifts vertically by -5.

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43 [Challenging] • Target: Logarithmic & Exponential Functions Practice Exercises • Cross-Lesson: 6.01 Permutations and Combinations

The function f is defined by $f(x) = x^2 - 6x + 13 \quad \text{for } x \geq 3$. Determine the complete set of real values of the constant k for which the equation $f^{-1}(x) = |2x - k|$ has exactly one real solution.

a. $k = \frac{39}{8}$ **or** $5 < k < 11$ ✓

b. $k = \frac{39}{8}$ or $5 \leq k < 11$

c. $\frac{39}{8} < k \leq 5$ or $k \geq 11$

d. $k = \frac{39}{8}$ only

1. **Find $f^{-1}(x)$ and its domain:** Completing the square gives $f(x) = (x - 3)^2 + 4$. For the domain $x \geq 3$, $f(x)$ is strictly increasing with range $[4, \infty)$. Setting $y = (x - 3)^2 + 4$ with $x \geq 3$ yields $x - 3 = \sqrt{y - 4}$, so $f^{-1}(x) = 3 + \sqrt{x - 4}, \quad \text{with domain } x \geq 4$. 2. **Set up the equation $f^{-1}(x) = |2x - k|$:** Let $u = \sqrt{x - 4} \geq 0$, which implies $x = u^2 + 4 \geq 4$. The equation becomes $3 + u = |2(u^2 + 4) - k| = |2u^2 + 8 - k|$. 3. **Analyz.

44 [Challenging] • Target: Logarithmic & Exponential Functions Practice Exercises • Cross-Lesson: 7.01 Differentiation and Rates of Change

Consider the function $f(x) = \frac{x^3 - 8}{x^2 - 4}$. We are interested in the behavior of this function near its discontinuities. Let $x = a$ be a value where the function has a removable discontinuity, and let $x = b$ be a value where the function has a non-removable discontinuity (vertical asymptote). What is the value of the limit of $f(x)$ as x approaches a, and what is the value of $\lim_{x \to b^-} f(x)$?

a. **Limit at a is 3,** $\lim_{x \to b^-} f(x) = -\infty$ ✓

b. Limit at a is 3, $\lim_{x \to b^-} f(x) = \infty$

c. Limit at a is 1, $\lim_{x \to b^-} f(x) = -\infty$

d. Limit at a is 1, $\lim_{x \to b^-} f(x) = \infty$

First, we factor the numerator and the denominator of $f(x) = \frac{x^3 - 8}{x^2 - 4}$. The numerator is a difference of cubes: $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$. The denominator is a difference of squares: $x^2 - 4 = (x - 2)(x + 2)$. Thus, $f(x) = \frac{(x - 2)(x^2 + 2x + 4)}{(x - 2)(x + 2)}$. A removable discontinuity occurs where a factor cancels out. Here, $(x - 2)$ cancels, so the removable discontinuity is at $x = a = 2$. To find the limit as x approaches a, we consider the simplified function for $xeq2$.

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45 [Challenging] • Target: Logarithmic & Exponential Functions Practice Exercises • Cross-Lesson: 8.01 Integration and Kinematics

The function f is defined for $x \in \mathbb{R}$ by $f(x) = 3x + k,$ where k is a constant. The function g is defined for $x \in \mathbb{R}$ by $g(x) = |x - 4|.$ Given that $f^{-1}(4) = 2$, find the value of k and all real solutions to the equation $f^{-1}(x) = g(x)$.

a. $k = -2;\ x = \frac{5}{2},\ x = 7$ ✓

b. $k = -2;\ x = 7$

c. $k = 2;\ x = \frac{7}{2},\ x = 5$

d. $k = -2;\ x = -\frac{5}{2},\ x = 7$

1. **Find the parameter k:** By the definition of an inverse function, $f^{-1}(4) = 2 \Leftrightarrow f(2) = 4$. Substituting $x = 2$ into $f(x)$: $f(2) = 3(2) + k = 6 + k\ 6 + k = 4 \Rightarrow k = -2$ 2. **Determine the inverse function $f^{-1}(x)$:** With $k = -2$, $f(x) = 3x - 2$. Setting $y = 3x - 2$ and solving for x: $3x = y + 2 \Rightarrow x = \frac{y+2}{3}$ Thus, $f^{-1}(x) = \frac{x+2}{3}$. 3. **Solve the equation $f^{-1}(x) = g(x)$:** $\frac{x+2}{3} = |x - 4| \Leftrightarrow x + 2 = 3|x - 4|$.

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